

Data Analysis with R and Python — Quiz 2 — May 7, 2025

NAME:

ROLL NUMBER:

Execute the following code to generate 100 random samples from the $\text{Beta}(2.5, 5)$ distribution.

```
set.seed(20252502)
x <- rbeta(100, 2.5, 5)
```

Assume that this data is a random sample $X_1, X_2, \dots, X_n \sim \text{i.i.d. Beta}(\alpha, \beta)$ for some unknown $\alpha, \beta > 0$.

- (a) Find the maximum likelihood estimates of α and β based on the generated data. Note that explicit expressions for the maximum likelihood estimators are *not* available for this parametric family. [6]

$$\hat{\alpha} = \qquad \qquad \qquad \hat{\beta} =$$

- (b) Briefly outline how you obtained these estimates (code is not required). [4]